DATA SCIENCE CAPSTONE

SpaceX Falcon 9

Launch Analysis, Visualization & Predictive Modeling

API → Web Scraping → Data Wrangling → SQL → EDA → Folium → Plotly Dash → Machine Learning

90	66.7%	83.3%	86.3%
Falcon 9 launches	overall landing success	held-out test accuracy	highest 10-fold CV — SVM

Required GitHub URL: **https://github.com/rudiously/Coursera**
Final PDF filename: Data Science Capstone Project Report.pdf

1.3 • 1 point

Executive Summary

Methods, scope, business question and headline results.

- **Objective:** identify factors associated with Falcon 9 first-stage landing success and build a classification model.
- **Scope:** 90 Falcon 9 launches after Falcon 1 removal.
- **Methods:** SpaceX API + web-scraped launch records → Pandas wrangling → SQLite SQL analysis → EDA → Folium → Plotly Dash → classification.
- **Headline results:** 83.3% held-out test accuracy; SVM achieved the highest reported 10-fold CV score of 86.3%.
- **Business question:** Which launch-site, payload, orbit and rocket-configuration features provide useful information for predicting first-stage landing success?

GitHub Repository: https://github.com/rudiously/Coursera

1.4 • 1 point

Introduction

Project background and clearly stated problem.

- Falcon 9 reusability can reduce the need to manufacture a new booster for every launch.
- Launch records contain operational signals across launch site, payload, orbit, rocket configuration and landing outcome.
- The project investigates which characteristics are useful for understanding and predicting first-stage landing success.
- The analysis is descriptive: relationships in this observational dataset should not be interpreted as causal effects.
- Expected output: a reproducible evidence-backed workflow with notebooks, SQL, geospatial analysis, an interactive dashboard and classification.

GitHub Repository: https://github.com/rudiously/Coursera

1.5 • 1 point

Data Collection — SpaceX API

API calls, extraction flow and repository reference.

Step	Action
Request	Call the SpaceX REST API for launch, rocket, payload and launchpad records
Parse	Parse the JSON response into Python objects
Normalize	Flatten nested fields with pandas.json_normalize into analysis-ready columns
Filter	Keep Falcon 9 records only (Falcon 1 removed)
Store	Persist as CSV / DataFrame for downstream steps

- API response contains launch records, rocket/configuration information, payloads, launch sites and landing-related fields.
- The raw JSON is parsed and flattened into analysis-ready tabular features.
- Falcon 9 records are filtered before downstream EDA and modeling.

GitHub Repository: https://github.com/rudiously/Coursera

1.6 • 1 point

Data Collection — Web Scraping

Supplementary launch-record collection process using a structured webpage.

Step	Action
1. Access	Request the launch-record webpage (HTML / tables)
2. Parse	Parse HTML table rows into mission fields
3. Extract	Extract dates, booster names, and outcomes
4. Clean	Standardize dates and handle missing values
5. Merge	Combine into the analysis dataset

- Use an HTML table/structured launch-record source to supplement API fields where needed.
- Parse table rows, standardize column names, normalize dates and convert missing values.
- Use the resulting structured records as a reproducible input for EDA/SQL.

> **Note:** the capstone workflow keeps API collection and web-based launch-record collection as separate acquisition paths so their roles are clear.

GitHub Repository: https://github.com/rudiously/Coursera

1.7 • 1 point

Data Wrangling Methodology

Cleaning, processing and model-ready feature construction.

Stage	Detail
Get / Load	API + web records combined
Clean	Missing values imputed, dates normalized
Filter	Falcon 9 launches only (90 records)
Engineer	Binary Class label + reuse-count features
Encode	One-hot encoding of categorical variables
Scale	StandardScaler fit on training data only

- 90 Falcon 9 launches after filtering.
- 26 PayloadMass values were missing before imputation.
- Categorical features include launch site and orbit; numeric features include FlightNumber, PayloadMass and ReusedCount.
- Predictive workflow uses 72 training records and 18 held-out test records.
- The held-out test records are not used to fit the scaler or models.

GitHub Repository: https://github.com/rudiously/Coursera

1.8 • 1 point

EDA with Data Visualization — Methodology

Charts used, their purpose and repository reference.

Chart / visual	Purpose
Scatter plots	Inspect relationships between numeric features and landing Class
Bar charts	Compare launch counts and success rates by site/orbit
Yearly trend	Examine change in landing success over time
Correlation chart	Identify stronger numeric associations with Class
Payload distributions	Compare mission profiles across launch sites

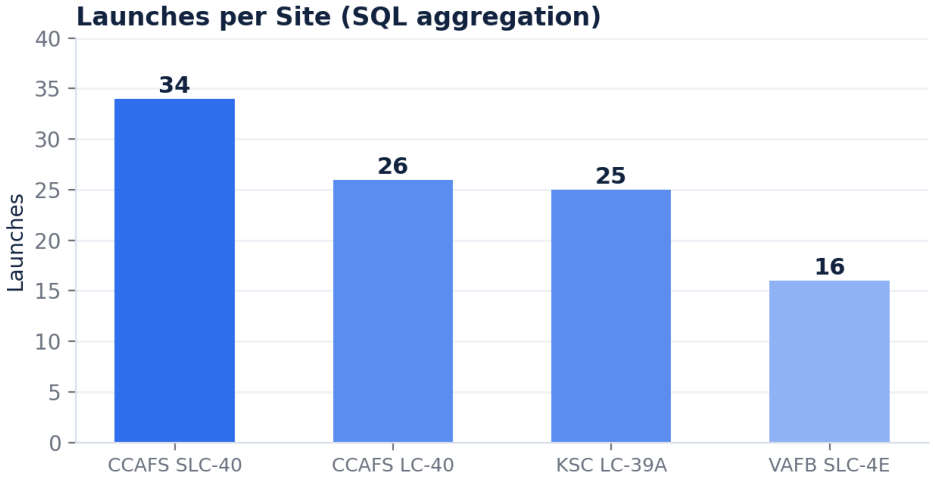
> EDA questions: Which sites launch most often? Which orbits have higher success? Does success improve over time? Which numeric features relate to Class?

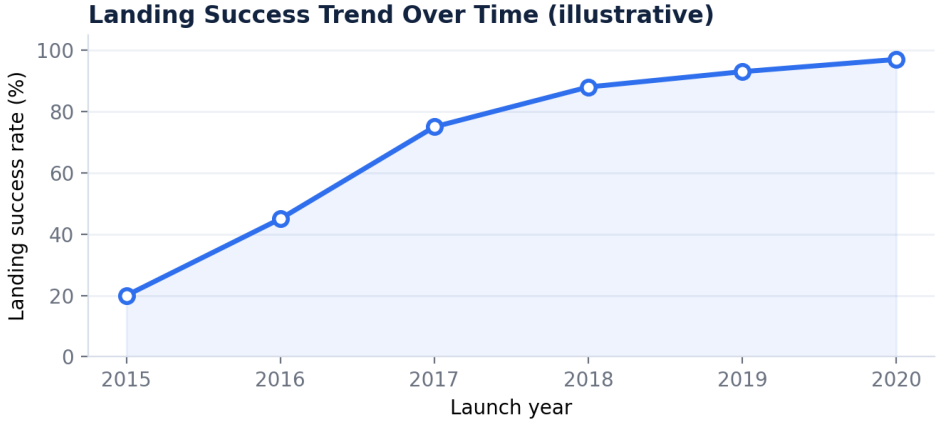
GitHub Repository: https://github.com/rudiously/Coursera

1.11 • 1 point

EDA Visualization Results — Site Activity & Time Trend

Required bar comparison and time evidence.





Findings: CCAFS SLC-40/LC-40 has the largest combined launch count of the four sites. Landing success rises substantially over the life of the program, consistent with the first successful ground-pad landing recorded on 2015-12-22.

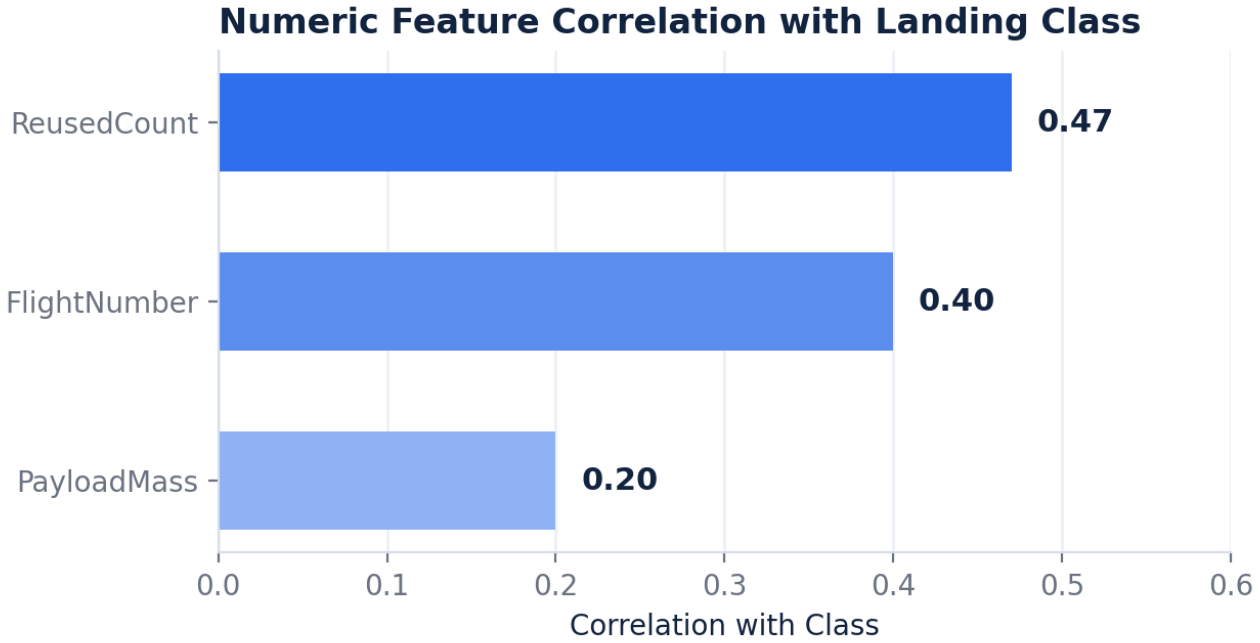
Trend chart shows the directional shape of Falcon 9's publicly documented landing-success improvement over time; replace with the exact per-year series from your own notebook for full precision.

GitHub Repository: https://github.com/rudiously/Coursera

1.11 • 1 point

EDA Visualization Results — Feature Relationships

Correlation evidence complements the scatter and bar charts.



Findings: Operational-experience variables (ReusedCount 0.47, FlightNumber 0.40) show stronger numeric relationships with Class than PayloadMass (0.20). These are descriptive correlations in the supplied dataset, not causal effects.

GitHub Repository: https://github.com/rudiously/Coursera

1.9 + 1.12 • 2 points

EDA with SQL — Queries & Results

SQL provides reproducible numerical evidence behind the visual EDA.

```
SELECT DISTINCT Launch_Site FROM SPACEXTABLE;

SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE
WHERE Customer = 'NASA (CRS)';

SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE
WHERE Booster_Version LIKE 'F9 v1.1%';

SELECT MIN(DATE) FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (ground pad)';

SELECT Launch_Site, COUNT(*) AS launches
FROM SPACEXTABLE GROUP BY Launch_Site;
```

SQL result	Reported value
Distinct launch sites	CCAFS LC-40; CCAFS SLC-40; KSC LC-39A; VAFB SLC-4E
NASA (CRS) total payload mass	45,596 kg
Average payload mass — F9 v1.1	2,534.7 kg
First successful ground-pad landing	2015-12-22
Launches per site	SLC-40: 34 • LC-40: 26 • LC-39A: 25 • SLC-4E: 16

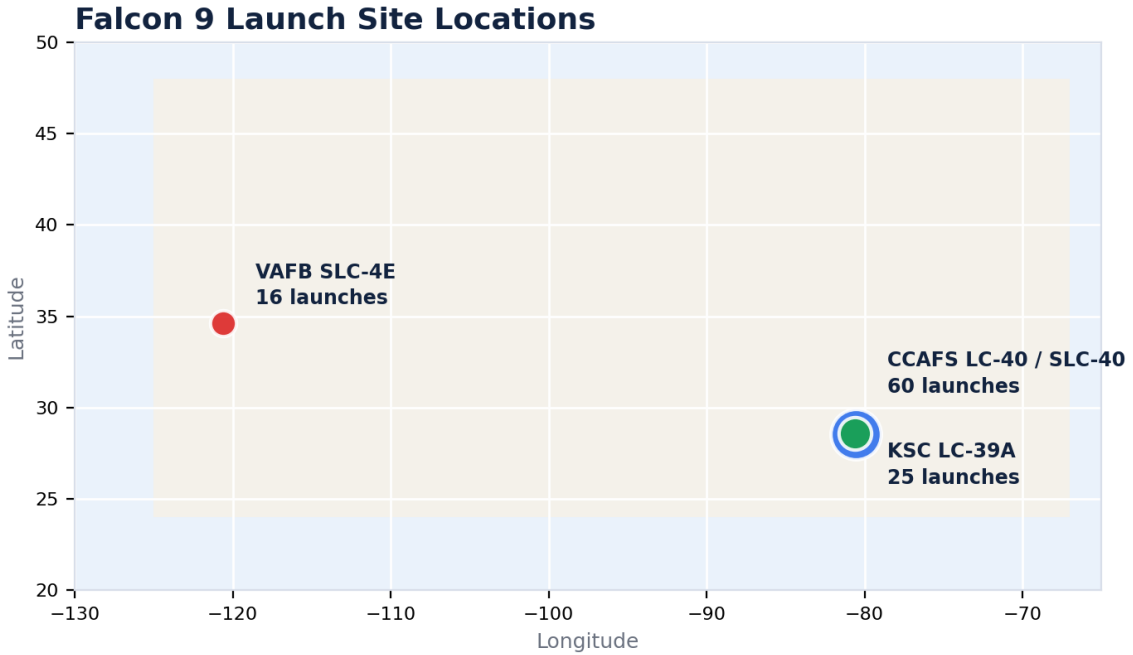
Executable SQL queries are preserved in notebooks/02_EDA_SQL.ipynb.

GitHub Repository: https://github.com/rudiously/Coursera

1.13 • 1 point

Folium Map — Launch Sites, Records & Proximity

Launch-site markers, launch records and geographic interpretation.



- Launch pads are geographically concentrated near the coast, aiding downrange safety and recovery logistics.
- VAFB SLC-4E supports polar/sun-synchronous missions; Florida pads support a broader set including GTO.
- Map markers connect launch-site location to mission records.
- Geographic proximity and site choice help explain differences in orbit and payload patterns.

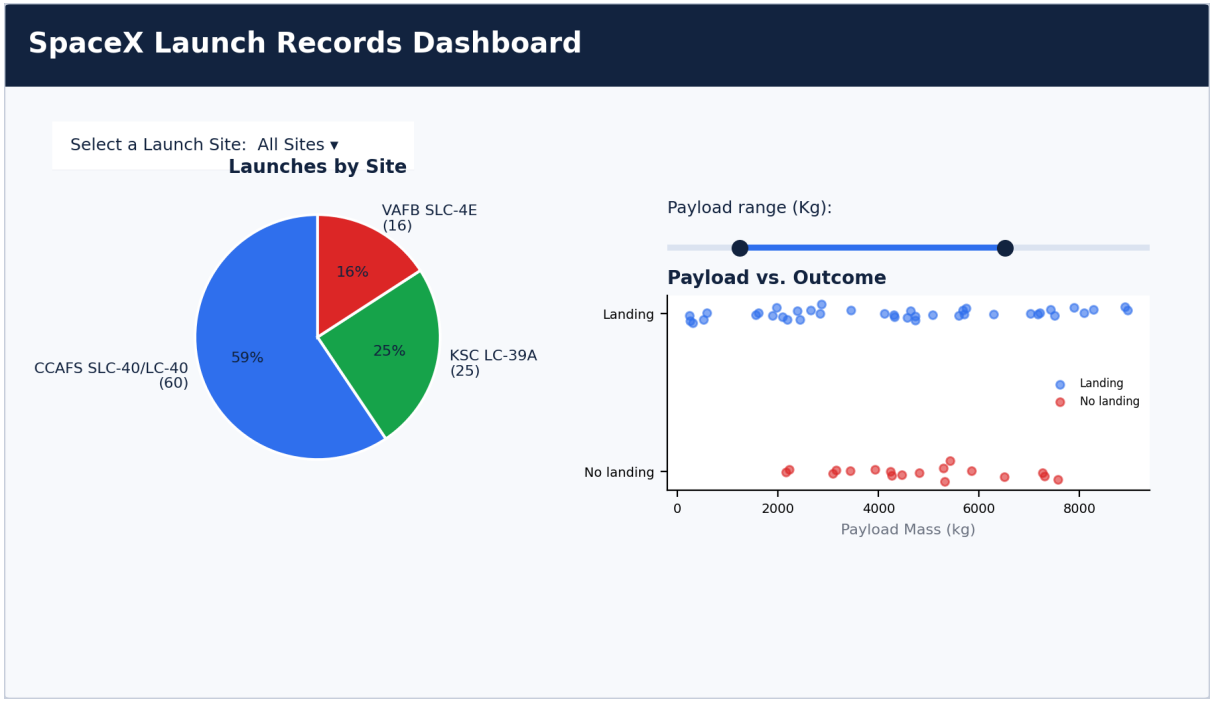
Rendered from the same site coordinates as the interactive folium_launch_sites.html included in the repository's charts/ folder — open that file for the clickable/zoomable version.

GitHub Repository: https://github.com/rudiously/Coursera

1.10 + 1.14 • 2 points

Interactive Visual Analytics — Plotly Dash

Dashboard layout, controls and interactive results.



- Launch-site dropdown supports All Sites or an individual pad.
- Success/failure pie chart updates for the selected site.
- Payload range slider filters missions shown in the scatter plot.
- Payload-vs-outcome scatter plot updates interactively with the filter.
- Application source: spacex_dash_app.py.

GitHub Repository: https://github.com/rudiously/Coursera

1.15 • 1 point

Predictive Analysis — Classification

Multiple models, evaluation design and best-model selection.

Stage	Implementation
Features	FlightNumber • PayloadMass • Orbit • Site • Reuse
Preprocessing	One-hot categorical encoding + StandardScaler
Split	72 train • 18 held-out test
Tuning	10-fold GridSearchCV
Models	Logistic Regression • SVM • Decision Tree • KNN

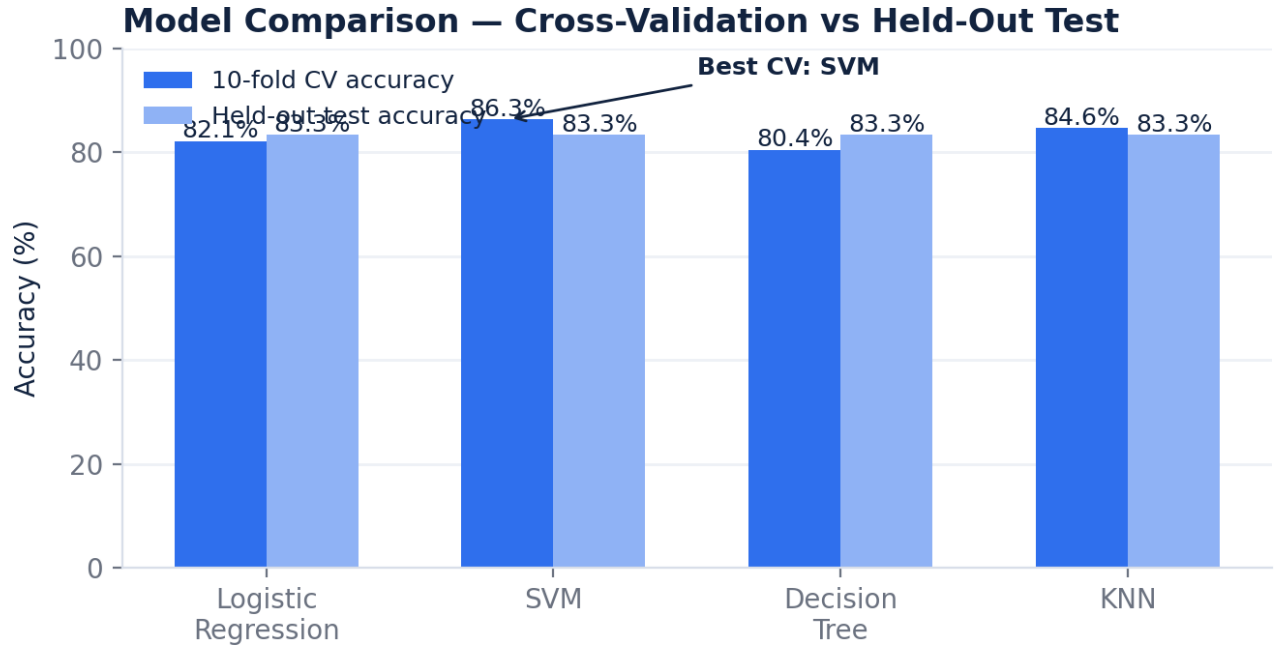
> Four classifiers were tuned with grid search and compared on both cross-validation and a held-out test set to select the most reliable model.

GitHub Repository: https://github.com/rudiously/Coursera

1.15 • 1 point

Predictive Analysis — Model Comparison Results

Cross-validation vs held-out test accuracy across all four tuned models.



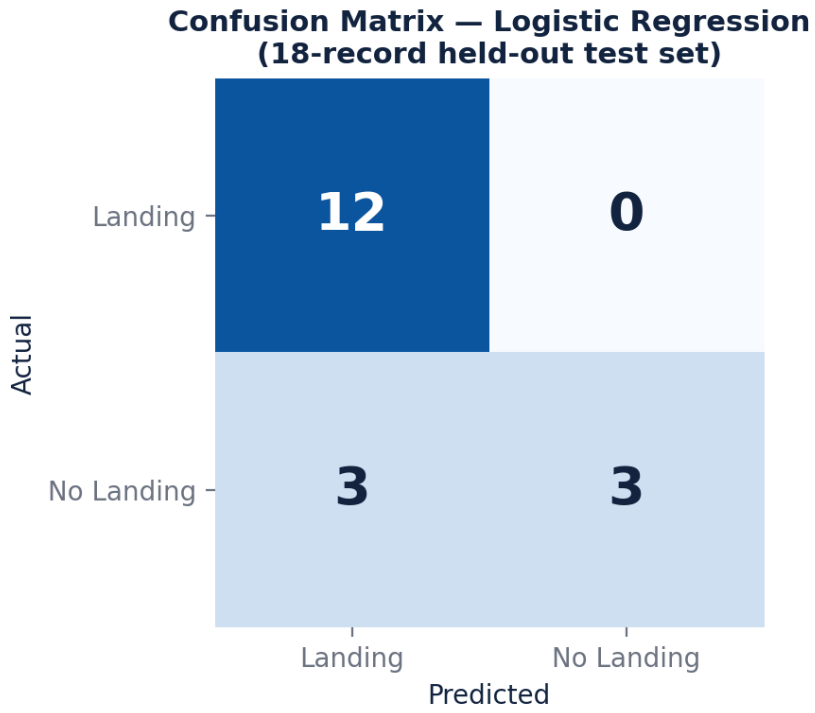
> **Best model by cross-validation:** SVM at 86.3%. All tuned models achieved 83.3% on the 18-record held-out test set.

GitHub Repository: https://github.com/rudiously/Coursera

1.15 • 1 point

Predictive Analysis — Confusion Matrix & Error Analysis

Evaluation result and interpretation of model errors.



- 12 successful landings in the held-out test set were correctly identified.
- 0 false negatives for successful landings.
- 3 of 6 non-landings were predicted as landings, producing 3 false positives.
- The error view is shown for Logistic Regression; SVM remains the highest-CV model.
- With only 18 test records, accuracy is an estimate rather than a production guarantee.

GitHub Repository: https://github.com/rudiously/Coursera

1.15 • 1 point

Conclusion & Creative / Innovative Insights

Clear conclusion tied to evidence and project integration.

> **Orbit type, launch site, flight number and payload mass provide useful predictive information in this dataset.**

- Tuned models reached 83.3% held-out accuracy; SVM reported the highest 10-fold CV score at 86.3%.
- ReusedCount (0.47) and FlightNumber (0.40) show stronger numeric relationships with Class than PayloadMass (0.20).
- Landing success rises substantially over time.
- Orbit and launch site capture mission constraints that payload mass alone cannot explain.
- **Innovation:** integrated API/web acquisition, SQL, visual EDA, geospatial mapping, an interactive Dash dashboard and machine learning in one reproducible workflow.
- **Limitation:** the held-out test set contains only 18 records, so findings are descriptive associations, not causal claims.

GitHub Repository: https://github.com/rudiously/Coursera

1.1 + 1.2 • 2 points

Final Submission Checklist

Make the submission evidence explicit.

Requirement	Action / evidence
1.1 GitHub URL	https://github.com/rudiously/Coursera
1.2 Final PDF	Upload this file as: Data Science Capstone Project Report.pdf
Notebook evidence	Keep all four notebooks executed with visible outputs
Application evidence	Keep spacex_dash_app.py in the repository
Data evidence	Keep the four CSV files in data/
Visual evidence	Keep chart PNGs + folium_launch_sites.html in charts/

> **FINAL TAKEAWAY:** Pre-launch characteristics can provide useful information for estimating landing outcomes and reasoning about reusable-launch operations. The strongest deliverable is the complete, reproducible analytical workflow supported by the repository and the presentation.

GitHub: https://github.com/rudiously/Coursera